Vivek Sai

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Education

M.Sc. Computer Science (Autonomous Systems) | *University of Stuttgart*, Stuttgart, Germany — Oct 2024 – Present

B.Tech. Computer Science & Engineering (AI) | *Amrita Vishwa Vidyapeetham*, Chennai, India — Oct 2020 – Apr 2024

Work Experience

Working Student – Humanoid Robotics, Development & Validation — Apr 2026 – Present

Porsche Engineering Services GmbH — *Stuttgart, Germany*

- Deployed and validated an NVIDIA cuVSLAM + nvblox perception-localization stack on Unitree Go2/G1 (Jetson Orin NX, RealSense D435i), resolving DDS, TF-tree, and domain-ID conflicts to sustain 30 Hz visual odometry and real-time 3D reconstruction.
- Unified Go2 and G1 onto a single arm64 YOLOv8/TensorRT detection pipeline (Isaac ROS), fixing build, QoS, and runtime defects to reach real-time detections at ∼24 Hz on embedded Jetson hardware.
- Designed a SOLID-refactored ROS 2 service converting 2D object detections into 3D poses at 30 Hz with full unit-test coverage, enabling downstream navigation and manipulation.
- Re-architected the robot audio stack: unified text-to-speech into a shared cross-robot scheduler and shipped a swappable ASR backend (Nemotron) with VAD-based turn detection and wake-word gating.

Student Assistant — Nov 2025 – Jun 2026

Socially Intelligent Robotics (SIR) Lab, University of Stuttgart — *Stuttgart, Germany*

- Debugged hardware–software integration issues on 2 FER robotic platforms, cutting experiment setup failures by ∼40% and improving uptime.
- Prototyped and evaluated learning-based robotic behaviors in NVIDIA Isaac Sim, focusing on sim-to-real transfer.

Student Assistant — Dec 2024 – Mar 2026

Construction Robotics, University of Stuttgart — *Stuttgart, Germany*

- Implemented and validated pose estimation and control loops, conducting sim-to-real validation that identified a ∼15% perception accuracy gap and reduced it to under 5% through calibration and pipeline tuning.
- Trained a PPO-based reinforcement learning policy for peg-in-hole insertion on a Franka arm in simulation, tuning reward shaping, action space, and hyperparameters to improve policy convergence.
- Trained a SmolVLA (VLA) model on a SO-100 arm in NVIDIA Isaac Sim, achieving 40–60% inference success.

Research Intern — Jun 2023 – Aug 2023

IIITDM Kancheepuram — *Chennai, India*

- Implemented and benchmarked deep learning models (CNNs, Vision Transformers) in PyTorch/TensorFlow, with data preprocessing, augmentation, and evaluation pipelines to analyze robustness and generalization.

Projects

ContactPilot — Vision-Based Robotic Grasping Pipeline | ***Python, PyTorch, MuJoCo, Open3D, SAM 3, LLMs***

- Built an end-to-end pick-and-place pipeline for a Franka Panda arm with two interchangeable 6-DoF grasp backends (Contact-GraspNet, NVIDIA GraspGen), multi-camera point-cloud fusion, and vision-only occupancy-based bin placement for collision-free packing.
- Designed a seed-based benchmarking harness with automated failure-taxonomy analysis, raising pick-and-place success from 38% to a sustained 100% (30/30 objects, 10-seed benchmark); added SAM 3 promptable object selection (mask IoU 0.29 → 0.89–0.99) and an LLM instruction parser (Llama 3.1 via NVIDIA NIM) for multi-step free-text commands.

Generalizable VLA Policies for Franka Manipulation | ***Python, MuJoCo, robosuite, HuggingFace***

- Designed a systematic multi-axis generalization study (spatial, language, distractors) fine-tuning SmolVLA and X-VLA on sealed, scripted datasets (24–560 episodes); published 7 datasets and 9 model checkpoints to Hugging Face.
- X-VLA reached 88.1% on unseen colour-instruction pairs (held-out within 10 pts of trained) and 94.4% on language grounding with no positional shortcut verified; identified and fixed a classifier bug via video inspection.

3D LiDAR Odometry (MAD-ICP) | ***C++, ROS 2, Eigen, Python***

- Implemented a MAD-ICP-inspired 3D LiDAR odometry system in C++ (Eigen-only core + ROS 2 Humble wrapper); source-voxel downsampling (0.15 m) gave ∼1.9× wall-clock speedup on 128-beam LiDAR, achieving ATE RMSE 0.014 m (handheld), RPE drift 2.8%/4.4% on SPOT loop/stair datasets.

Skills

Programming:	C++, Python, Linux/Bash
Robotics:	ROS 2, robot kinematics, motion execution, path planning, localization, sim-to-real transfer
Perception:	Intel RealSense D435i, NVIDIA cuVSLAM, nvblox, camera calibration, 6D pose estimation (AprilTag)
Simulation:	NVIDIA Isaac Sim, MuJoCo, Gazebo, URDF
ML / Deep Learning:	PyTorch, CNNs, Vision Transformers, reinforcement learning (PPO), VLA fine-tuning (SmolVLA, X-VLA), model evaluation
Hardware:	Unitree Go2 (quadruped), Unitree G1 (humanoid), Jetson Orin NX, FER robotic platforms, SO-100
Tools:	Git, Docker, unitree_sdk2
Languages:	English (C1), German (A1)